

University of Technology Sydney

42050 SAS Predictive Business Analytics

Assessment Task 1

Data Exploration

**UCI Bank Marketing: Improving Term-Deposit Campaign
Targeting**

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Dataset: UCI Bank Marketing - bank.csv

1. Executive Summary and Business Problem

Purpose. This report examines the UCI Bank Marketing data in SAS Viya to identify customer and campaign characteristics associated with term-deposit subscription, assess data quality, and create a defensible preparation workflow for later propensity modelling.

The work addresses Subject Learning Objectives 1 and 2 by defining a business analytics problem and applying SAS Viya data profiling, preparation and visual analysis to support evidence-based decisions.

Business problem. Telephone campaigns consume staff time and can create customer fatigue. The practical goal is therefore not simply to predict y, but to support better targeting: direct contact effort toward higher-propensity groups, avoid unnecessary repeat contact, and preserve a clear distinction between information available before a call and information known only afterward.

Key findings

- The target is imbalanced: 521 subscriptions (11.52%) versus 4,000 non-subscriptions (88.48%). Accuracy alone would therefore be a weak later modelling criterion.
- There are no technical numeric missing values, but semantic unknown levels are material: contact is unknown for 1,324 records (29.29%) and poutcome for 3,705 records (81.95%).
- Previous campaign outcome is the strongest signal encountered. Customers with poutcome=success subscribed at 64.34%, compared with 9.10% for unknown.
- Housing and personal loans are associated with lower conversion: 8.60% for housing-loan customers versus 15.34% without, and 6.22% for personal-loan customers versus 12.48% without.
- Retired (23.48%) and student (22.62%) customers show the highest substantial job-group conversion rates; blue-collar customers show the lowest among large groups (7.29%).
- The duration variable is descriptively informative but was rejected for pre-call modelling because call duration is unavailable before the call ends and would create leakage.

Recommended action. Prioritise customers with positive previous-campaign evidence and no current loan burden; test retired and student segments with controlled outreach. Use uplift or holdout testing before operational rollout because the analysis is observational, not causal.

2. Dataset Characteristics

The analysis uses the UCI Bank Marketing bank.csv file, a 10% sample of the older 17-variable dataset. It contains 4,521 observations, 16 candidate predictors, and the binary target y. The classification context, sample size, mixed data types, and interpretable campaign variables make the data suitable for this assessment.

Group	Variables	SAS role / treatment
Customer	age, job, marital, education	age interval; categorical variables nominal
Financial	default, balance, housing, loan	balance interval; yes/no fields binary
Current campaign	contact, day, month, duration, campaign	day/campaign interval; contact/month nominal; duration rejected for modelling
Campaign history	pdays, previous, poutcome	pdays/previous interval; poutcome nominal
Outcome	y	Binary target; event level=yes

Numeric profile

Variable	Min	Mean	SD	Max	Interpretation
age	19	41.17	10.58	87	moderate spread
balance	-3,313	1,422.66	3,009.64	71,188	strong right tail; negative values valid
day	1	15.92	8.25	31	bounded calendar value
duration	4	263.96	259.86	3,025	right-skewed; leakage risk
campaign	1	2.79	3.11	50	repeated-contact tail
pdays	-1	39.77	100.12	871	-1 means never contacted
previous	0	0.54	1.69	25	zero-heavy

Target and evaluation implication

A naive all-no classifier would already be 88.48% accurate. Later predictive work should therefore emphasise event-sensitive measures such as recall, precision, lift, ROC/AUC, and misclassification cost rather than accuracy in isolation. The Model Studio partition was stratified to preserve the event rate across training, validation, and testing samples.

3. Data Profiling and Validation

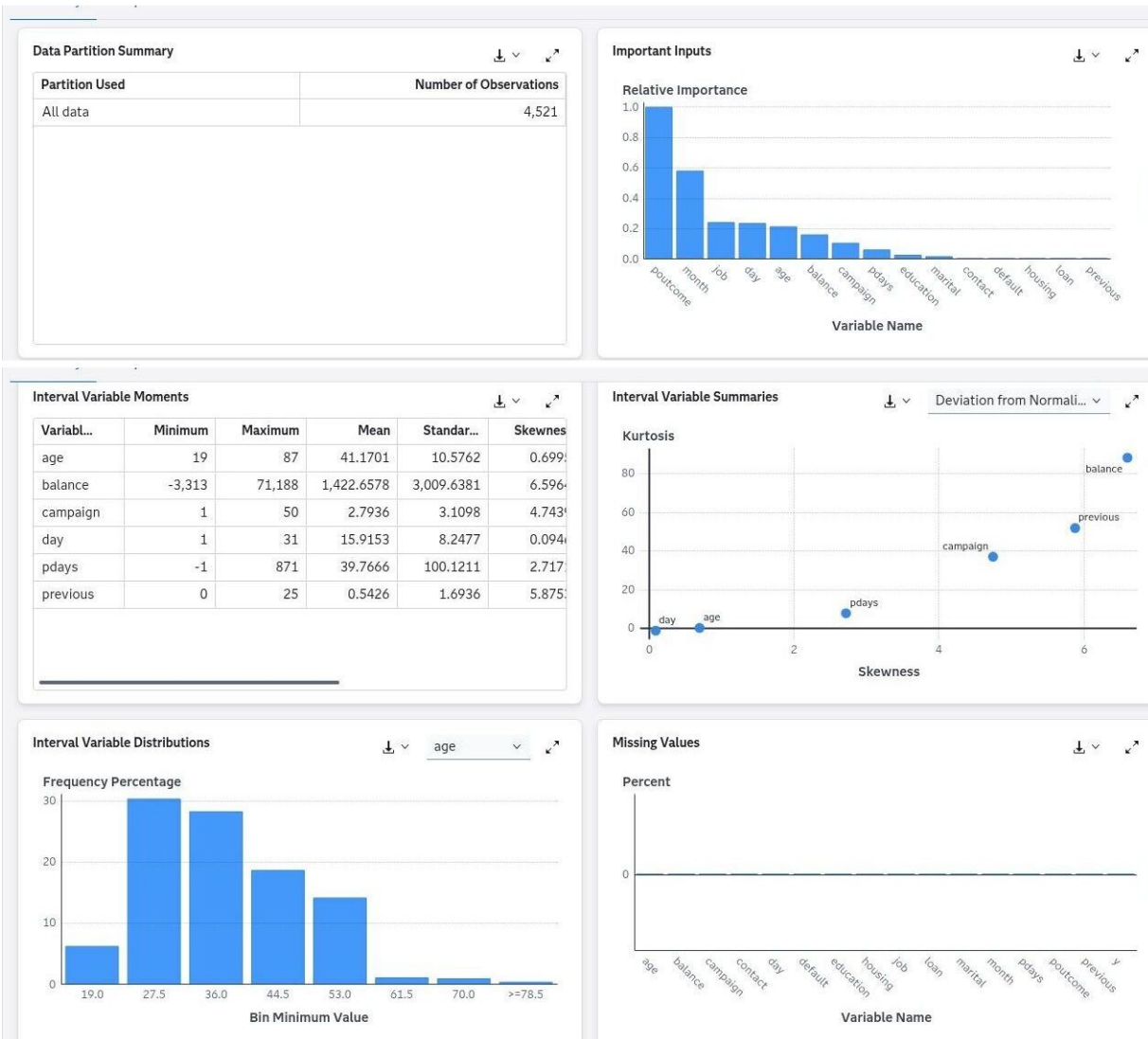


Figure 1. SAS Model Studio Data Exploration: Important Inputs, interval moments, distribution diagnostics and missing-value profile.

Data Exploration used all 4,521 observations. Tree-split importance ranked poutcome first (relative importance 1.0000), followed by month (0.5807) and job (0.2417). This is evidence of predictive association and screening usefulness, not causality; it supports retaining campaign history, timing and customer segment for later modelling.

The interval profile shows pronounced non-normality for balance (skewness 6.5964; kurtosis 88.3903), campaign (4.7439; 37.1689), pdays (2.7171; 7.9571) and previous (5.8753; 51.9952). Every input has zero technical missing values. The preparation decision was therefore controlled 3-SD capping in Replacement, while preserving valid negative balance, pdays=-1 and zero previous contacts; no blanket missing-value conversion was warranted.

Correlation evidence. Pearson and mutual-information heat maps were produced for redundancy screening. No unsupported coefficient or causal inference is reported.

4. Data Quality, Roles and Partitioning

The screenshot shows the SAS Model Studio interface for a project named 'A1_Bank_Marketing_Maurya'. The 'Data' tab is active, displaying a table of variables. The variable 'y' is selected, and its metadata is shown on the right. The 'y' variable is defined as a binary target with an event level of 'yes' and is rejected from modeling.

Variable Name	Type	Role	Assess for Bias	Level
age	Numeric	Input		Interval
balance	Numeric	Input		Interval
campaign	Numeric	Input		Interval
contact	Character	Input		Nominal
day	Numeric	Input		Interval
default	Character	Input		Binary
duration	Numeric	Rejected		Interval
education	Character	Input		Nominal
housing	Character	Input		Binary
job	Character	Input		Nominal
loan	Character	Input		Binary
marital	Character	Input		Nominal
month	Character	Input		Nominal
pdays	Numeric	Input		Interval
poutcome	Character	Input		Nominal
previous	Numeric	Input		Interval
y	Character	Target		Binary

Metadata for 'y':

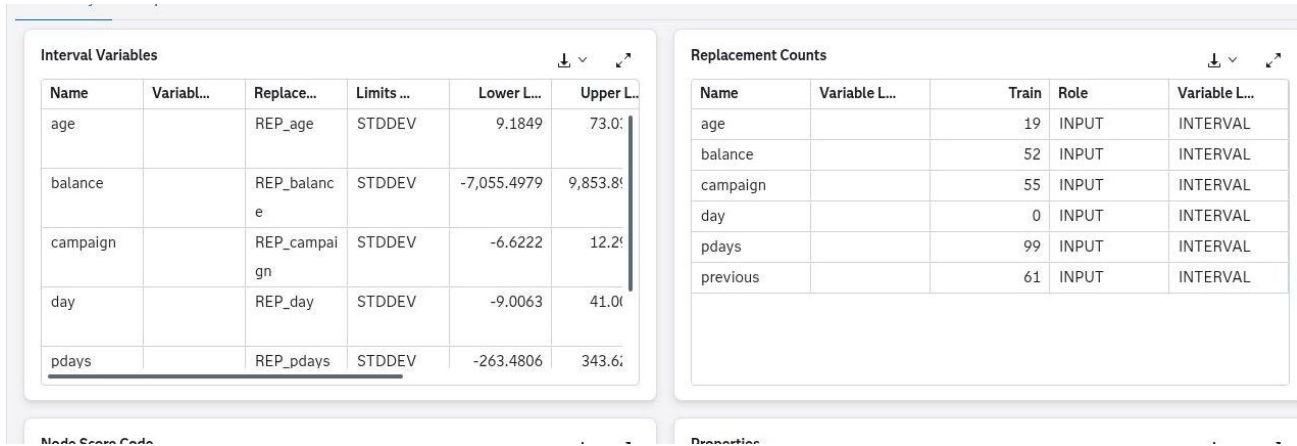
- Role: Target
- Level: Binary
- Specify the Target Event Level: yes
- Order: Default
- Transform: (empty)
- Impute: (empty)

Figure 2. Model Studio metadata: y defined as a binary target with event level yes; duration rejected.

Issue	Decision	Reason
Target event	y=yes	Aligns the model with the subscription business outcome.
Class imbalance	Stratified 60/30/10 partition	Maintains comparable event prevalence in train, validation and test data.
Event-based sampling	Off	Avoids distorting prevalence for this small assignment dataset.
duration	Rejected from modelling	Known only after the call; using it for pre-call targeting would leak future information.
unknown categories	Retained as explicit levels	They may carry operational meaning and should be investigated, not silently imputed.
pdays=-1	Retained	Represents no prior contact rather than a numeric error.

An independent check of the validated CSV found zero exact duplicate rows and no SAS-missing source values, so duplicate removal and row deletion were not justified. The workflow preserves valid business states and documents leakage controls at metadata level.

5. Transformation and Feature-Engineering Workflow



Name	Variabl...	Replace...	Limits ...	Lower L...	Upper L...
age		REP_age	STDDEV	9.1849	73.0
balance		REP_balanc e	STDDEV	-7,055.4979	9,853.8
campaign		REP_campai gn	STDDEV	-6.6222	12.2
day		REP_day	STDDEV	-9.0063	41.0
pdays		REP_pdays	STDDEV	-263.4806	343.6

Name	Variable L...	Train	Role	Variable L...
age		19	INPUT	INTERVAL
balance		52	INPUT	INTERVAL
campaign		55	INPUT	INTERVAL
day		0	INPUT	INTERVAL
pdays		99	INPUT	INTERVAL
previous		61	INPUT	INTERVAL

Figure 3. Replacement results: computed 3-SD limits and training-partition replacement counts.

Node	Exact variables affected	Exact SAS setting / method	Rationale	Result
Replacement	age, balance, campaign, day, pdays, previous → REP_ fields	Standard deviations from mean; 3 SD; computed limits; unknown class replacement=Ignore.	Control extreme tails without redefining valid business states.	Training replacements: 19, 52, 55, 0, 99, 61 respectively. Negative balance and pdays=-1 remain within limits.
Transformations	No transformed variables created; node received class inputs and REP_interval fields.	Default interval=(none); default class=(none); missing treatment=Separate; reject originals ON.	Avoid unsupported transformations where none was selected.	Node executed; results contain input statistics only and no transformation outputs.
Imputation	All downstream inputs; each has 0 missing and 0 imputable values.	Impute non-missing OFF; cutoff 50%; class=Count; interval=Mean; limits=All data (5%).	Retain a reproducible downstream stage without treating unknown or pdays=-1 as missing.	No material replacements occurred.
Variable Selection	contact, housing, job, loan, marital, month, poutcome	Fast Supervised ON; Unsupervised ON; Linear Regression ON; combination = selected by at least 1.	Combine complementary statistical screens.	Seven inputs retained.
Variable Clustering	Class inputs plus replacement numeric fields	Best variable; include class inputs ON; export class indicators OFF; rho=0.80.	Reduce redundant inputs and improve interpretability.	Three clusters; representatives poutcome_unknown, contact_unknown and education_tertiary.

Reproducibility. The executed pipeline was saved to Exchange as A1_Bank_Marketing_Maurya_Preparation_Model. Live node properties and results confirm the settings above. These point-and-click settings are not recreated by the SAS program. The text-mining branch was not used because no text-role field was available.

6. Variable Selection

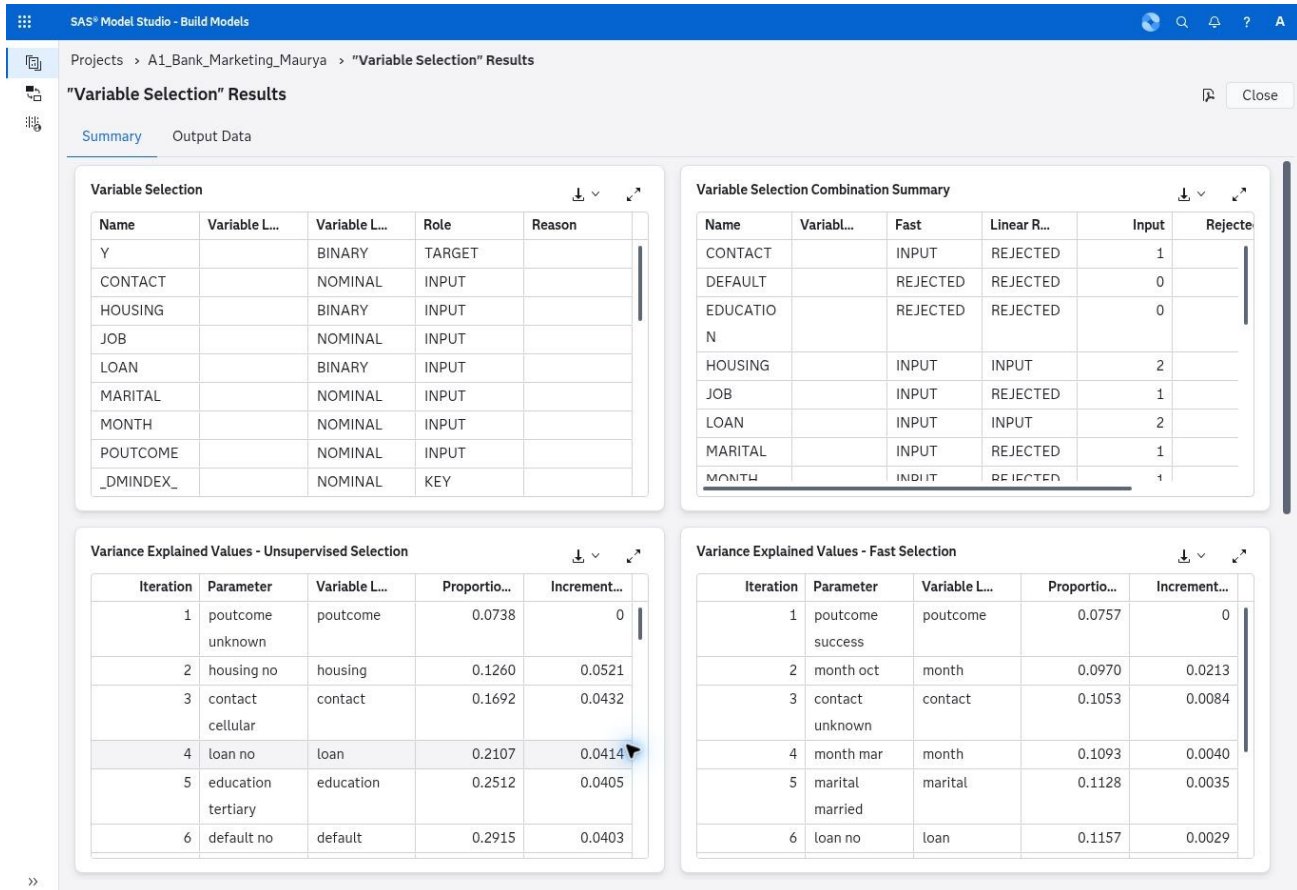


Figure 4. SAS Model Studio Variable Selection results and retained-input summary.

The node combined Fast Supervised Selection, Unsupervised Selection and Linear Regression, with variables accepted when selected by at least one method. The retained inputs were contact, housing, job, loan, marital, month and poutcome.

Evidence	Interpretation
poutcome selected by both supervised and regression screens	Historical campaign outcome carries the clearest direct subscription signal.
housing and loan selected by both screens	Existing debt commitments are consistently associated with lower subscription propensity.
contact, job, marital and month selected by the fast screen	Operational timing and customer-segment information adds non-linear or level-specific separation.
poutcome_success estimate ≈ 0.538 ; $p < .0001$	Successful prior response is the dominant positive indicator in the linear screening results.

Caution. Selection indicates statistical usefulness in this sample; it does not establish causality. Rare levels such as month=dec and job=unknown should not drive policy without confidence intervals or a larger validation sample.

7. Variable Clustering and Redundancy Control

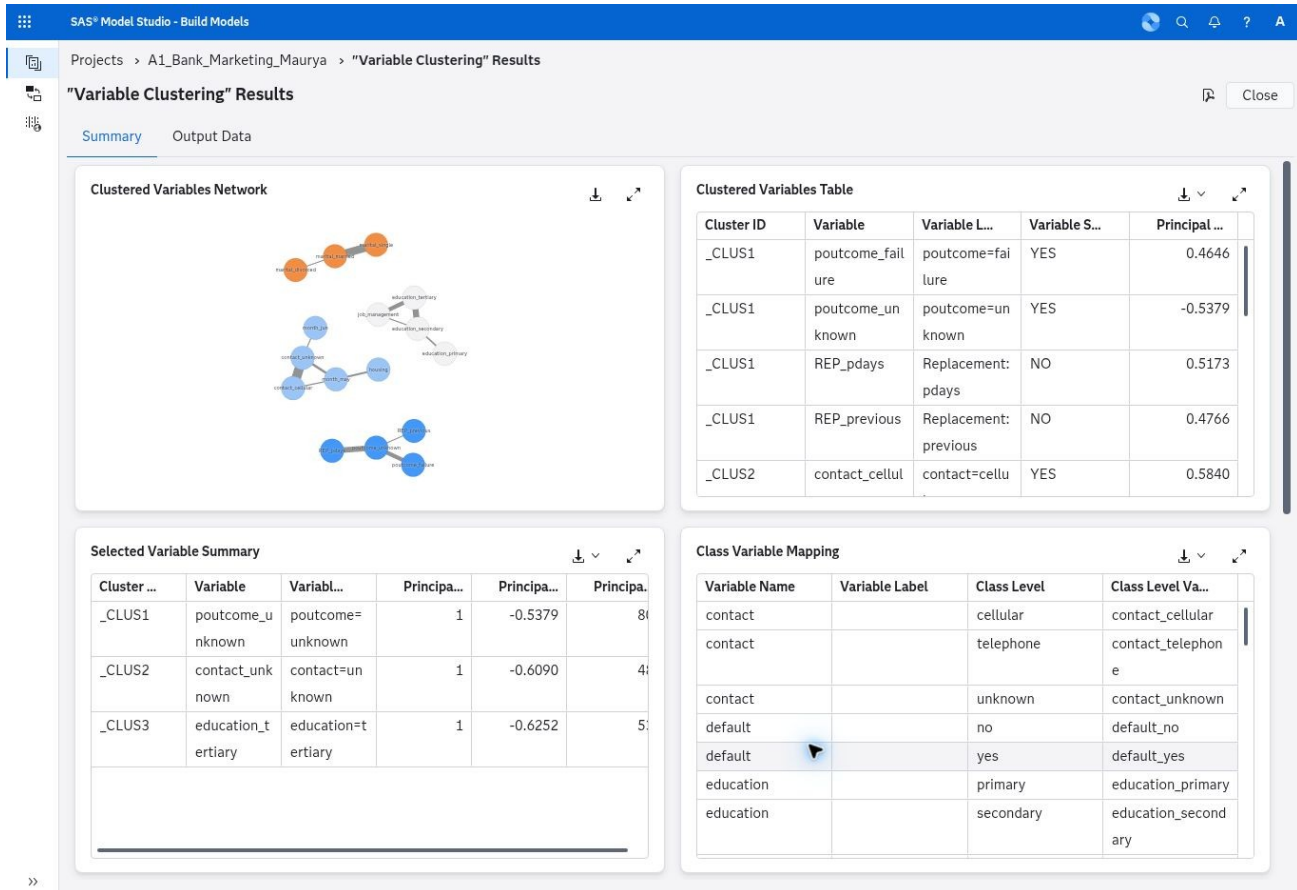


Figure 5. Variable Clustering output using best-variable representation and class inputs.

The clustering node used a rho threshold of 0.80, included class inputs, and represented each cluster by its best variable. It produced three interpretable clusters: campaign-history state, contact/channel and housing/month pattern, and education indicators.

Cluster	Members with strongest structure	Representative
1 — campaign history	poutcome_failure, poutcome_unknown, REP_pdays, REP_previous	poutcome_unknown
2 — contact / campaign context	contact_cellular, contact_unknown, housing, month_jun, month_may	contact_unknown
3 — education	education_primary, education_secondary, education_tertiary	education_tertiary

This step prevents the feature space from being dominated by redundant indicators. The selected representatives are statistical summaries of clusters; business interpretation should still refer to the underlying variables and levels.

8. Visual Analytics: Previous Campaign Outcome

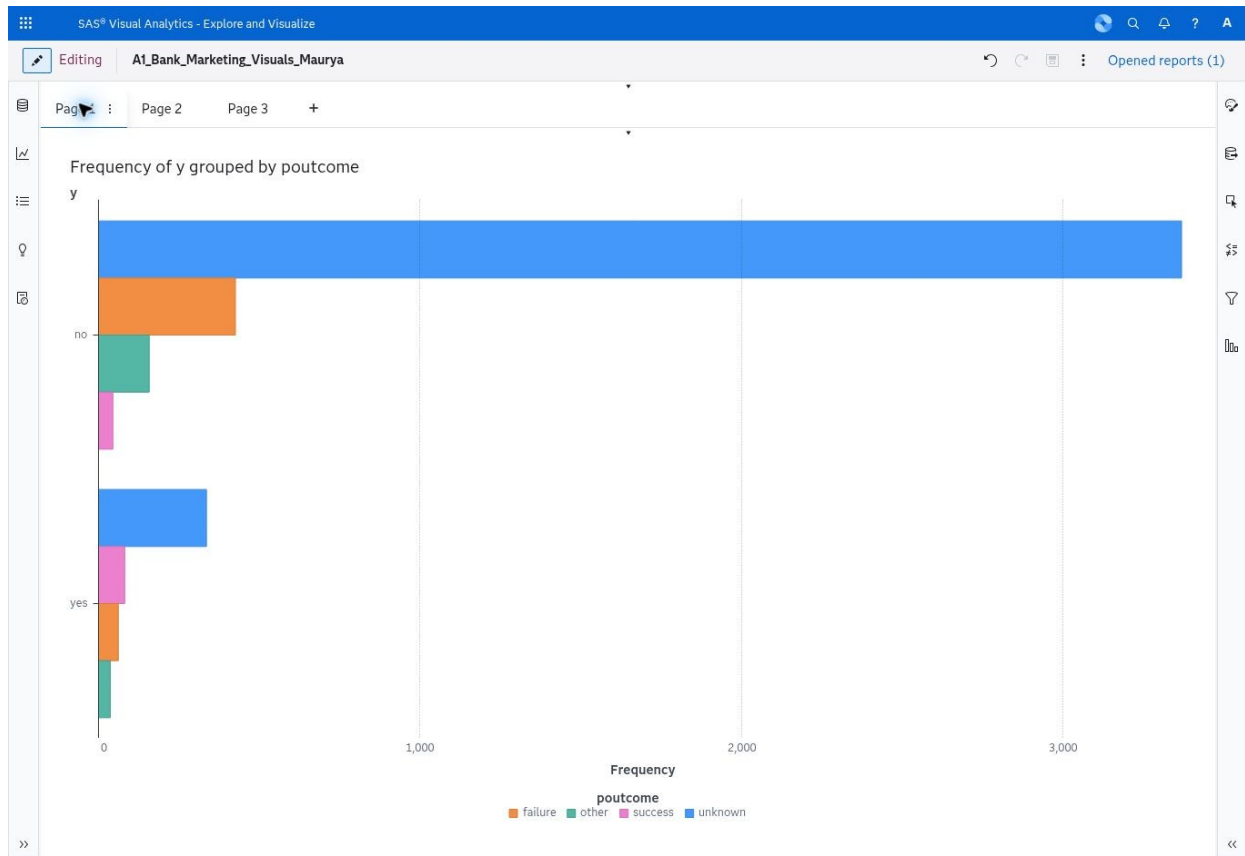


Figure 6. Saved SAS Visual Analytics page: subscription outcome grouped by previous campaign outcome.

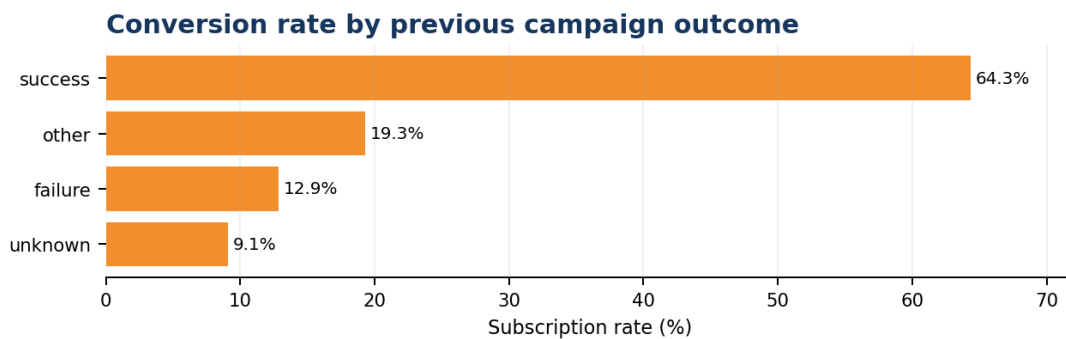


Figure 7. Conversion-rate view calculated from the validated 4,521-row table.

The frequency view shows that unknown dominates the sample, so raw counts alone can obscure effectiveness. Rate calculations reveal the business signal: success converts at 64.34% (83/129), other at 19.29% (38/197), failure at 12.86% (63/490), and unknown at 9.10% (337/3,705). A previous success is therefore a strong prioritisation cue, while the large unknown group remains the main volume opportunity.

9. Visual Analytics: Customer Segment

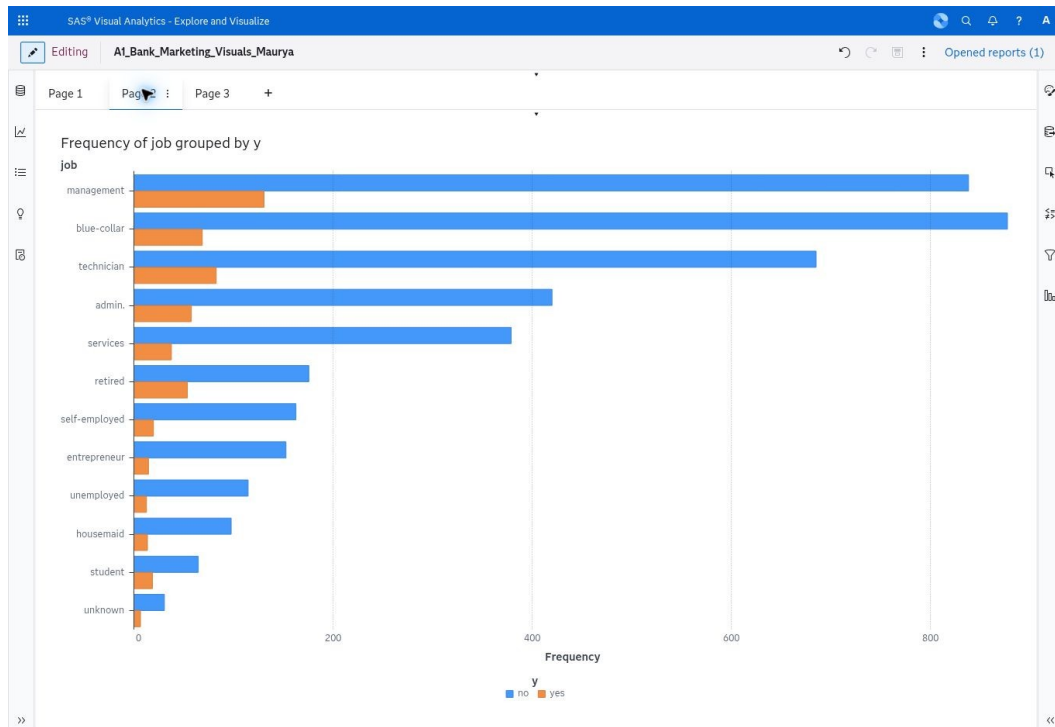


Figure 8. Saved SAS Visual Analytics page: job frequency grouped by subscription outcome.

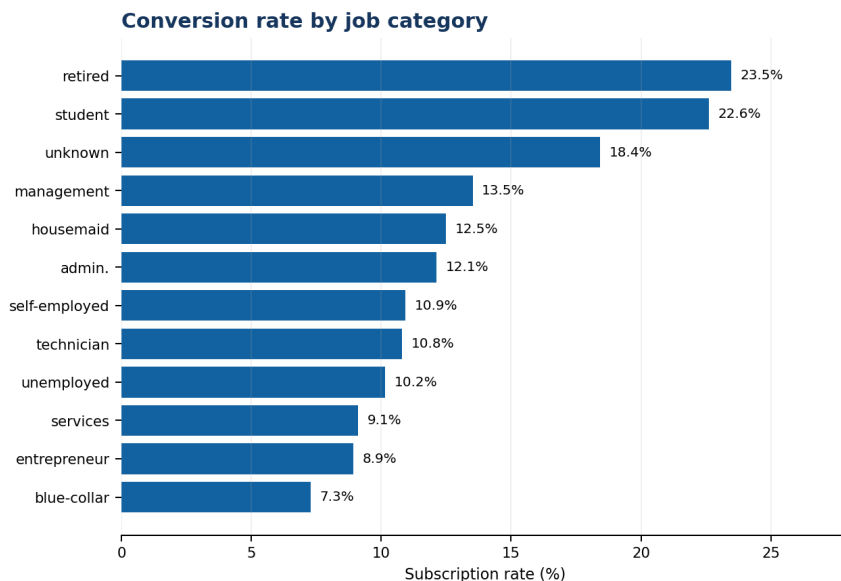


Figure 9. Subscription rates by job category.

Management and blue-collar categories contribute large volumes, but their rates differ materially: 13.52% versus 7.29%. Retired (23.48%) and student (22.62%) customers have the highest substantial group rates. The unknown job group is small ($n=38$), so its 18.42% rate is unstable and should not be overinterpreted.

Business implication. Segment contact strategy by both expected rate and reachable volume. Retired and student segments merit targeted testing; management provides a larger scalable pool; blue-collar outreach should be more selective.

10. Visual Analytics: Campaign Timing

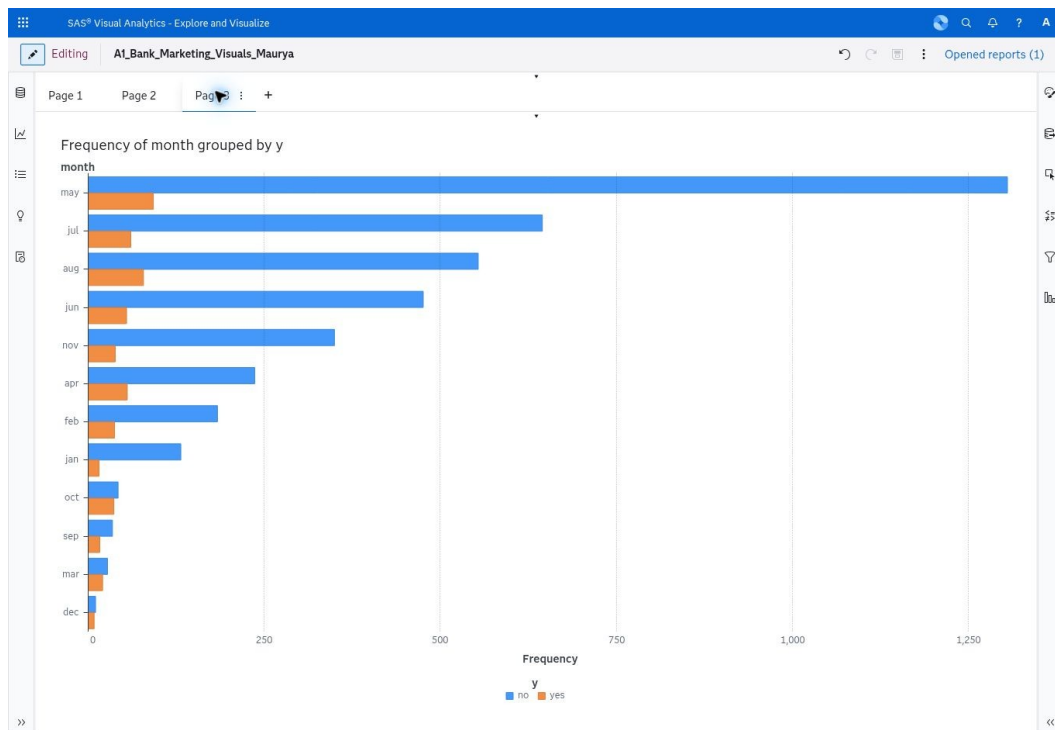


Figure 10. Saved SAS Visual Analytics page: contact month grouped by subscription outcome.

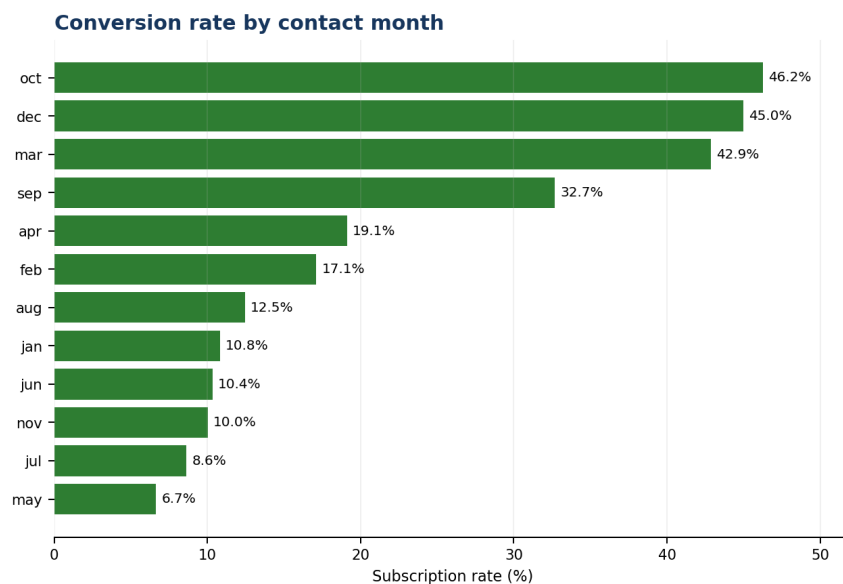
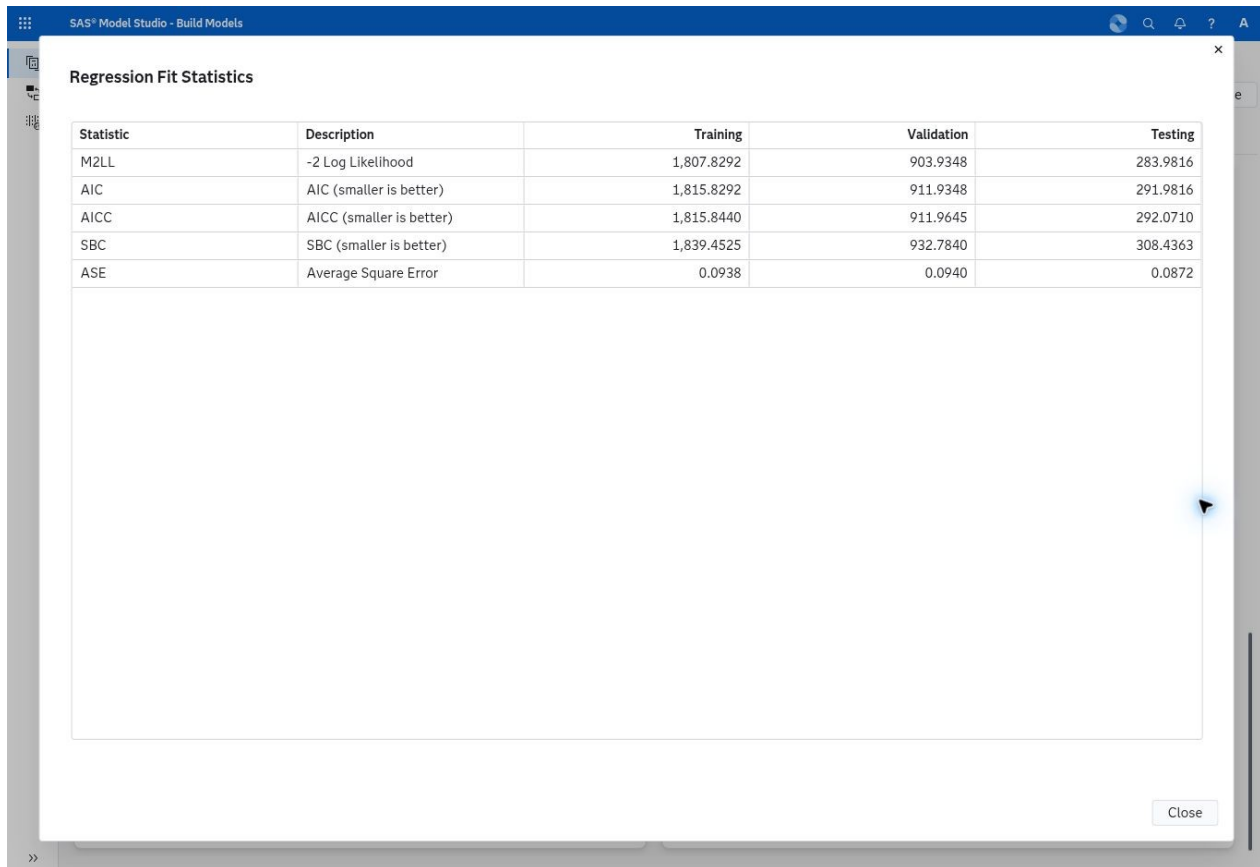


Figure 11. Subscription rate by contact month.

May contains the greatest contact volume but the lowest conversion rate (6.65%). October (46.25%), December (45.00%), March (42.86%) and September (32.69%) appear much stronger, but their sample sizes are only 80, 20, 49 and 52 respectively. This may reflect campaign design or customer-selection effects rather than a pure calendar effect.

Business implication. Use month as a planning and model input, but validate rare-month effects across additional years before reallocating campaign capacity.

11. Optional Logistic Modelling Extension



The screenshot shows the 'Regression Fit Statistics' window in SAS Model Studio. It contains a table with the following data:

Statistic	Description	Training	Validation	Testing
M2LL	-2 Log Likelihood	1,807.8292	903.9348	283.9816
AIC	AIC (smaller is better)	1,815.8292	911.9348	291.9816
AICC	AICC (smaller is better)	1,815.8440	911.9645	292.0710
SBC	SBC (smaller is better)	1,839.4525	932.7840	308.4363
ASE	Average Square Error	0.0938	0.0940	0.0872

Figure 12. Stepwise logistic-regression fit statistics in SAS Model Studio.

A leakage-controlled logistic model was run as an extension to the required preparation work. Stepwise SBC retained only poutcome. The selected model reduced training SBC from 1,948.23 (intercept only) to 1,753.66 and achieved assessment squared error of 0.09385, 0.09403 and 0.08723 on training, validation and testing respectively.

Parameter	Estimate	p-value	Interpretation vs unknown
Intercept	-2.281	<.0001	Low baseline subscription odds
poutcome=failure	0.315	0.1011	Not statistically distinct at 5%
poutcome=other	0.844	0.0004	Higher odds than unknown
poutcome=success	2.873	<.0001	Very strong positive association

Scope limitation. This compact model is evidence of the dominance of campaign history, not a production-ready targeting model. Threshold choice, lift, calibration, fairness, operational cost and temporal validation remain necessary before deployment.

12. Recommendations, Limitations and Contribution Log

Recommendations

- Prioritise previous campaign success and customers without housing or personal loans, subject to contact-policy and consent constraints.
- Use job and month as segmentation variables, but combine conversion rate with segment size and uncertainty.
- Retain unknown as an explicit operational state and investigate why contact and previous-outcome information is often absent.
- Exclude duration from pre-call scoring; evaluate future models with event-sensitive and cost-sensitive measures.
- Run a controlled holdout or A/B test before claiming that any segment or timing choice causes improved conversion.

Limitations

The file is a 10% sample of an older campaign dataset and may not represent current customers or channels. The analysis is observational, some high-rate levels are rare, and no customer-contact costs are supplied. Semantic unknowns are frequent, while the bank's campaign-selection process is not documented; both factors may confound comparisons.

Individual contribution log

Contributor	Specific tasks / sections completed	Challenges addressed
Aditya Maurya -	Dataset acquisition and CAS import; SAS Studio validation; metadata, target/event and partition setup; Data Exploration; preparation pipeline; Variable Selection and Clustering; Visual Analytics; optional logistic extension; interpretation and report assembly.	Detected duration leakage; preserved pdays=-1 and negative balances; retained semantic unknown levels; managed the 11.52% event class; reduced redundant inputs; saved the pipeline for reproducibility.

References

Moro, S., Rita, P., & Cortez, P. (2014). Bank Marketing [Dataset]. UCI Machine Learning Repository. <https://doi.org/10.24432/C5K306>

SAS Institute Inc. SAS Viya applications used: SAS Studio, SAS Model Studio, and SAS Visual Analytics (course environment, accessed 1 September 2026).

Saved SAS artifacts. Project: A1_Bank_Marketing_Maurya. Public table: BANK_MARKETING_A1_MAURYA. Exchange pipeline: A1_Bank_Marketing_Maurya_Preparation_Model. Visual report: A1_Bank_Marketing_Visuals_Maurya.

Generative AI use disclosure. OpenAI ChatGPT assisted with drafting, language editing, formatting and consistency checks. Reported results, SAS settings and screenshots were verified against the supplied data and genuine SAS Viya project.